Pochak OTT Platform — Project Definition Document

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1. System Architecture

```
┌──────────────────────────────────────────────────────────────────────────┐
│                          CLIENTS (Frontend)                              │
│                                                                          │
│  ┌──────────────┐   ┌──────────────┐   ┌──────────────┐                 │
│  │  BO Web      │   │  Public Web  │   │  Mobile App  │                 │
│  │  Next.js 16  │   │  React+Vite  │   │  RN + Expo   │                 │
│  │  :3000       │   │  :3100       │   │  :8097       │                 │
│  │  65 pages    │   │  50 pages    │   │  59 screens  │                 │
│  └──────┬───────┘   └──────┬───────┘   └──────┬───────┘                 │
│         │                  │                  │                          │
│         └──────────────────┴──────────────────┘                          │
│  ┌──────────────────────────────────────────────────┐                    │
│  │              Shared Packages (15 files)          │                    │
│  │  @pochak/api-client · @pochak/domain-types       │                    │
│  │  @pochak/design-tokens · @pochak/utils           │                    │
│  └──────────────────────────────────────────────────┘                    │
└──────────────────────────────────────────────────────────────────────────┘
                                │
                                │ HTTPS / JWT
                                ▼
┌──────────────────────────────────────────────────────────────────────────┐
│                         API GATEWAY (:8080)                              │
│                      Spring Cloud Gateway (Reactive)                     │
│                                                                          │
│   Filters: JWT Validation → Redis Rate Limit → Correlation ID            │
│   Public: /home, /contents, /search, /sports, /schedule, /communities    │
│   Admin:  /admin/** (BO 전용)                                            │
└──────────────────────────────────────────────────────────────────────────┘
     │          │          │          │          │
     ▼          ▼          ▼          ▼          ▼
┌─────────┐┌─────────┐┌─────────┐┌─────────┐┌─────────┐
│Identity ││Content  ││Commerce ││Operation││ Admin   │
│ :8081   ││ :8082   ││ :8083   ││ :8084   ││ :8085   │
│ 66 Java ││241 Java ││ 69 Java ││ 60 Java ││103 Java │
└─────────┘└─────────┘└─────────┘└─────────┘└─────────┘
     │          │          │          │          │
     └──────────┴──────────┴──────────┴──────────┘
                                │
                     ┌──────────┴──────────┐
                     ▼                     ▼
          ┌──────────────┐        ┌──────────────┐
          │ PostgreSQL 16│        │   Redis 7    │
          │   :5432      │        │   :6379      │
          │  5 schemas   │        │  Rate Limit  │
          │ 18 migrations│        │  Cache       │
          └──────────────┘        └──────────────┘
```

2. Service Inventory

2.1 Backend Services

Service	Port	Schema	Java Files	Key Domains
pochak-gateway	8080	—	11	JWT, Rate Limit, CORS, Routing
pochak-identity-service	8081	identity	66	Auth, Signup (4경로), OAuth2, User Profile
pochak-content-service	8082	content	241	Organization (DAG), Competition, Match, Live/VOD/Clip, Tag, Membership, ACL, Search, Recommendation, Community, Home, Follow, Comment, Like, Favorite, Schedule, Highlight, Notification, Upload, Streaming
pochak-commerce-service	8083	commerce	69	Product, Purchase, Wallet, Coupon, Refund, Entitlement, GiftBall
pochak-operation-service	8084	operation	60	Venue, Camera, Reservation, Studio, Streaming Ingest, VPU
pochak-admin-service	8085	admin	103	RBAC, Analytics, Banner, Notice, Popup, Event, Terms, Inquiry, Report, AppVersion, AuditLog
pochak-common-lib	—	—	14	ApiResponse, ErrorCode, UserContext, DomainEvent, PageMeta

2.2 Frontend Applications

App	Framework	Port	Pages/Screens
BO Web	Next.js 16 (App Router)	3000	65 pages
Public Web	React 19 + Vite 8 (SPA)	3100	50 pages
Mobile App	React Native 0.76 + Expo 52	8097	59 screens

2.3 Infrastructure

Component	Version	Port	Purpose
PostgreSQL	16-alpine	5432	Primary DB (5 schemas)
Redis	7-alpine	6379	Rate Limiting, Cache
Docker Compose	—	—	10 containers

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3. Database Schema Map

```
PostgreSQL (pochak)
├── identity ─── users, user_auth_accounts, user_consents, user_preferences,
│                user_status_history, user_refresh_tokens, user_push_tokens,
│                user_notification_settings, user_relations,
│                phone_verifications, email_verifications
│
├── content ──── sports, sport_tags, organizations, teams, memberships,
│                competitions, competition_visits, matches, match_participants,
│                live_assets, vod_assets, clip_assets, asset_tags,
│                video_acl, view_history, display_sections, favorites,
│                watch_reservations, notifications, notification_recipients,
│                comments, follows, highlights, community_posts
│
├── commerce ─── products, purchases, wallets, wallet_ledger,
```

```
│                entitlements, refunds, coupons, user_coupons, gift_balls
│
├── operation ── venues, cameras, venue_cameras, reservations, studio_sessions
│
└── admin ────── admin_users, admin_roles, admin_groups, admin_menus,
                 admin_functions, admin_group_users, admin_group_roles,
                 admin_role_menus, admin_role_functions, audit_logs,
                 banners, popups, notices, events, advertisements,
                 inquiries, inquiry_answers, reports, terms,
                 app_versions, system_config, analytics_events
```

Migration History (18 files):

Version	Name	Tables
V000	init_schemas	5 schemas
V001	identity_schema	11 tables
V002	content_schema	~20 tables
V003	commerce_schema	6 tables
V004	operation_schema	6 tables
V005	admin_schema	15+ tables
V006	org_simplification_abac	4 tables
V007	coupon_tables	2 tables
V008	social_features	3 tables
V009	highlights	1 table
V010	analytics	1 table
V011	signup_account_system	3 tables
V012	fix_identity_schema	fixes
V013	**organization_v2_fields**	6 columns
V014	**competition_visibility**	2 columns + 1 table
V015	**community_posts**	1 table
V016	**cleanup_deprecated**	data migration
V100-V101	seed data	initial data

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4. Platform Areas (Policy v2)

4.1 포착 TV

```
메인 진입점 → 개인화 피드 (추천/인기/최신)
├── 홈 배너 (BO 관리)
├── 라이브 경기
├── VOD 다시보기
├── 클립 하이라이트
├── 대회/리그 목록
└── 검색/추천
```

4.2 포착 시티

```
지역 기반 체육시설 → 예약 현황
├── 시설 목록 (시/도/시/군/구 필터)
├── 시설 상세 (예약현황, 영업시간, 연락처)
├── 인증된 단체 (is_verified=true, BO 인증)
├── 촬영 예약 (ALL_MEMBERS 정책)
└── 공개 콘텐츠 (기본 PUBLIC)
```

4.3 포착 클럽

```
B2B/B2C 단체 → 가입 기반 접근
├── 오픈 단체: 자유 가입 + 공개 콘텐츠
├── CUG 단체: 단체 정보 공개 + 콘텐츠 제한
│   ├── 홍보용 영상 (visibility=PUBLIC, 비회원 시청 가능)
│   └── 회원 전용 영상 (visibility=MEMBERS_ONLY, 가입 후 시청)
├── 커뮤니티 (소식/구인/모집, 지역 기반)
└── 가입 정책: OPEN / APPROVAL / INVITE_ONLY
```

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5. API Endpoint Map

5.1 Identity Service (:8081)

Endpoint	Method	Description
`/auth/login`	POST	로그인
`/auth/signup`	POST	국내 성인 가입
`/auth/signup/minor`	POST	미성년 가입
`/auth/signup/social`	POST	SNS 가입
`/auth/signup/foreign`	POST	해외 가입
`/auth/phone/send-code`	POST	SMS 인증코드 발송
`/auth/phone/verify-code`	POST	인증코드 검증
`/auth/oauth2/callback/{provider}`	GET	OAuth2 콜백
`/auth/oauth2/complete-signup`	POST	OAuth 가입 완료
`/auth/refresh`	POST	토큰 갱신
`/users/me`	GET/PUT	프로필 조회/수정
`/users/me/preferences`	GET/PUT	선호 설정
`/admin/members`	GET	관리자 회원 목록

5.2 Content Service (:8082)

Endpoint	Method	Description
`/sports`	CRUD	종목 관리
`/organizations`	CRUD	**단체 관리 (DAG, display_area, CUG, join_policy)**
`/competitions`	CRUD	**대회 관리 (visibility, inviteCode)**
`/competitions/access`	POST	**비공개 대회 접근 (inviteCode 검증)**

Endpoint	Method	Description
`/competitions/visited`	GET	**방문한 비공개 대회 목록**
`/matches`	CRUD	경기 관리
`/contents/live`	CRUD	라이브 에셋
`/contents/vod`	CRUD	VOD 에셋
`/contents/clips`	CRUD	클립 에셋
`/contents/tags`	CRD	태그 관리
`/contents/{type}/{id}/access`	GET	**콘텐츠 접근 권한 확인 (CUG 포함)**
`/memberships`	CRUD	멤버십 관리 (승인/거절)
`/communities/posts`	CRUD	**커뮤니티 게시물 (소식/구인/모집/자유)**
`/communities/posts/{id}/pin`	PUT	게시물 고정
`/communities/posts/{id}/report`	POST	게시물 신고
`/home`	GET	홈 데이터
`/search`	GET	통합 검색
`/recommendations`	GET	추천 콘텐츠
`/follows`	POST/DELETE	팔로우
`/notifications`	CRUD	알림

5.3 Commerce Service (:8083)

Endpoint	Method	Description
`/products`	CRUD	상품 관리
`/purchases`	POST/GET	구매
`/wallet`	GET	지갑 잔액
`/wallet/charge`	POST	포인트 충전
`/wallet/use`	POST	포인트 사용
`/coupons`	GET/POST	쿠폰
`/refunds`	POST/GET/PUT	환불
`/entitlements/check`	GET	시청 권한 확인

5.4 Operation Service (:8084)

Endpoint	Method	Description
`/venues`	CRUD	경기장 관리
`/venues/search`	GET	경기장 검색
`/venues/nearby`	GET	주변 경기장
`/cameras`	CRUD	카메라 관리
`/reservations`	CRUD	촬영 예약
`/reservations/calendar`	GET	캘린더 뷰

Endpoint	Method	Description
`/streaming/ingest/*`	CRUD	스트리밍 인제스트
`/studio/sessions`	GET	스튜디오 세션

5.5 Admin Service (:8085)

Endpoint	Method	Description
`/admin/api/v1/auth/login`	POST	BO 로그인
`/admin/api/v1/rbac/*`	CRUD	RBAC 관리
`/admin/api/v1/analytics/*`	GET/POST	분석
`/admin/api/v1/site/*`	CRUD	배너/공지/팝업/이벤트
`/admin/api/v1/cs/*`	GET	고객센터
`/admin/api/v1/app/*`	CRUD	앱 버전

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6. BO Web Pages (65 pages)

Menu Structure (19 sections)

```
 1. 종목 관리 ──────────── /sports/list
 2. 협회 관리 ──────────── /teams/associations
 3. 구장 관리 ──────────── /venues/list
 4. 팀/단체 관리 ─────────── /teams/elite, club, private-hq, private-branch, public-org
 5. 대회/리그 관리 ──────── /competitions/list, schedule
 6. VPU 관리 ─────────────── /equipment/cameras, vpu-devices, vpu-contracts
 7. 촬영예약 관리 ─────────── /reservations, reservations/booking
 8. 스튜디오 관리 ─────────── /studio
 9. 컨텐츠 관리 ──────────── /contents/live, vod, clips, tags (+ create, upload)
10. 회원 관리 ──────────── /members/list, blacklist
11. 커뮤니티 관리 ─────────── /community/posts, reports        ★ Policy v2 신규
12. 뽈/시즌권/환불 ──────── /commerce/* (9개 서브메뉴)
13. 사이트 관리 ──────────── /site/popups, banners, notices, events
14. 앱 관리 ──────────────── /app-management
15. 고객센터 관리 ─────────── /support/inquiries, reports, terms
16. 운영 관리 ──────────── /operations/members, groups, permissions, menus
17. 모니터링 ──────────────── /monitoring
18. 통계 ──────────────── /statistics/users, views, sales
19. 스카이라이프 계약 ──────── /skylife/activation, vpu-chu
```

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7. Key Policy v2 Implementations

7.1 Organization DAG (단체 자가참조 계층)

```
// Organization.java
@ManyToOne(fetch = FetchType.LAZY)
@JoinColumn(name = "parent_id")
private Organization parent;  // 자가참조 DAG

// 신규 필드
private DisplayArea displayArea;      // CITY | CLUB
```

```
private Boolean isVerified;           // 시티 인증
private Boolean isCug;                // CUG 모드
private JoinPolicy joinPolicy;       // OPEN | APPROVAL | INVITE_ONLY
private ReservationPolicy reservationPolicy;  // ALL_MEMBERS | MANAGER_ONLY
private String siGunGuCode;           // 지역 코드
```

안전성: 순환참조 감지 + 최대 5단계 깊이 제한 (OrganizationService)

7.2 Competition Visibility (비공개 대회)

```
// Competition.java
private CompetitionVisibility visibility;  // PUBLIC | PRIVATE
private String inviteCode;                 // UUID (PRIVATE 시 자동 생성)

// CompetitionVisit.java — 방문 기록
@Table(uniqueConstraints = @UniqueConstraint({"user_id", "competition_id"}))
private Long userId, competitionId;
private LocalDateTime firstVisitedAt;
```

7.3 CUG Content Access Control

```
VideoAclService.evaluateAccess():
  1. CUG 단체 확인 (org.isCug)
  2. visibility=PUBLIC → ALLOW (홍보용)
  3. visibility=MEMBERS_ONLY → 멤버십 확인
  4. 비회원 → "가입 후 시청 가능" DENY
```

7.4 Community Module

```
// CommunityPost.java
PostType: NEWS | RECRUITING | RECRUITMENT | FREE
siGunGuCode: 지역 기반 노출
isPinned: 관리자 고정

// API: /communities/posts
GET (list + filter), POST, PUT, DELETE, PUT pin/unpin, POST report
```

---

8. Design System (CLAS Motif)

Token	Value
Primary	`#2563eb` (blue-600)
Primary Light	`#dbeafe` (blue-100)
Background	`#f8f9fb`
Card	`#ffffff` , border `#e5e7eb` , rounded-lg, shadow-sm
Sidebar	White, 220px, active=blue-50 + left 3px blue border
Header	White, 56px, bottom border
Font	Pretendard
Status Chips	완료=blue, 대기=amber, 실패=red, 활성=green

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9. Environment Variables

Variable	Service	Required
JWT_SECRET	Gateway, Identity	Yes (min 32 chars)
CORS_ALLOWED_ORIGINS	Gateway	Yes
DB_HOST/PORT/NAME/USER/PASSWORD	All services	Yes
REDIS_HOST/PORT	Gateway, Content	Yes
KAKAO_REST_API_KEY	Identity	Yes
GOOGLE_CLIENT_ID	Identity	Yes
NAVER_CLIENT_ID	Identity	Yes
COMMERCE_SERVICE_URL	Content	Default: localhost:8083
CDN_BASE_URL	Content, Operation	Default: cdn.pochak.co.kr

10. Build & Run

```
# Full stack (10 containers)
make all-up

# Development
make deps-up        # DB + Redis only
make bo-dev         # BO Web dev server (:3000)
make web-dev        # Public Web dev server (:3100)
make mobile-dev     # Expo Metro (:8097)

# Verify
make status         # Container status
make logs           # All logs
```

11. File Counts Summary

Component	Files	Lines (est.)
Backend Java	550	~25,000
BO Web (TSX/TS)	~120	~15,000
Public Web (TSX/TS)	~80	~8,000
Mobile (TSX/TS)	~90	~10,000
SQL Migrations	18	~1,600
Shared Packages	15	~500
Docker/CI/Infra	10	~300
Total	**~883**	**~60,000**

End of Project Definition Document